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SEX DIFFERENCES IN SPATIAL ABILITY: AN EVOLUTIONARY HYPOTHESIS AND TEST

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Although sex differences in cognitive functioning have been studied from many perspectives, current explanations consistently appeal only to proximate causes. An extensive literature details the developmental, neurological, hormonal, genetic, and experiential bases of cognitive sex differences (see, e.g., Maccoby and Jacklin 1974; Harris 1978; McGee 1979; Wittig and Petersen 1979), but virtually no attempt has been made to investigate the evolutionary basis. In other words, we have made considerable progress in learning how such sex differences emerge, but we cannot say why they occur. This paper focuses on one of the best-studied cognitive sex differences—the differential ability to solve spatial problems—and combines field and laboratory techniques to test the evolutionary hypothesis that such differences are a consequence of sexual selection for particular ranging patterns.

More than 50 years of psychological testing and research have yielded the consensus that "the most persistent of individual differences on multifactor tests of psychological functioning is a sex difference in spatial ability. Males have decidedly better spatial skill than females." (Harris 1978, p. 405; see additional reviews in Maccoby and Jacklin 1974; McGee 1979, 1982; Wittig and Petersen 1979.) In a detailed meta-analysis of the studies of cognitive sex differences, Rosenthal and Rubin (1982) concluded that, although the magnitude of such differences has declined in recent years, sex differences in cognitive functioning are still nontrivial and of real practical importance. The few nonhuman species for which there are comparative data suggest the generality of this phenomenon: among both wild and laboratory rodents males perform significantly better than females on spatial tasks (see, e.g., Hubbert 1915; Sadownikova-Koltzova 1926; McNemar and Stone 1932; Barrett and Ray 1970; Dawson 1972; Stewart et al. 1975; Joseph et al. 1978).

Spatial ability in humans is probably a collection of competencies subsumed under the single title because of their common reference to the visual-spatial

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domain. It is measured by a variety of tests requiring the subject to imagine the rotation of a depicted object, the folding of a pattern, or relative changes in the position of objects (e.g., Guilford and Lacey 1947); to recognize specified shapes when these are embedded in a more complex array (e.g., Gross 1959); to read a map (e.g., Money et al. 1965); or to solve a maze problem (e.g., Porteus 1965). Although factor-analytic studies suggest that these tests measure two or perhaps more subfactors of spatial ability (e.g., "visualization," "orientation," "familiar field"), men perform significantly better than women on all subfactors (Harris 1978; McGee 1979, 1982; Linn and Pulos 1983). In nonhuman subjects many of these measures of spatial ability are not applicable, and thus research has focused on maze-learning ability. Here also, consistent male superiority has been demonstrated (Buffery and Gray 1972; Dawson 1972; Stewart et al. 1975; Joseph et al. 1978).

HYPOTHESIS

Gray and Buffery offered a possible evolutionary explanation for the superior performance of males on spatial-learning tasks: "It seems likely that, among rodents, this male superiority is connected with the fact that, in the wild, this sex has a larger home range than does the female . . . [and that the] activity which is reflected in measurements of home range clearly involves the exercise of spatial ability" (Gray and Buffery 1971, p. 95). Their analysis permits two apparently different interpretations, and these are reflected in the supposedly competing "experiential" and "biological" models that dominate the spatial-ability literature (e.g., Wittig and Petersen 1979). The first view, which stresses ontogenetic factors, suggests that males develop superior spatial ability because their wanderlust gives them more experience in processing spatial data. The second view, which is evolutionary, holds that males perform better on spatial tasks because natural selection for such cognitive skill has acted more strongly on the sex that normally ranged farther. Gray and Buffery recognized that these views present a false dichotomy, simply explaining the phenomenon at different levels. Moreover, even when combined, these views provide an incomplete analysis, since the male's greater wanderlust or larger range clearly has some prior cause that requires explanation. We avoid this counterproductive opposition of biological and experiential causes and, instead, suggest an ultimate cause that can explain why males in certain species are expected to be both more mobile and more responsive to practice with spatial problems.

The critical fact Gray and Buffery (1971) failed to emphasize is that males do not universally have larger ranges. In subsequent work on sexual-selection theory Trivers (1972) offered an explanation for interspecific differences in the relative range size of the two sexes. Although no comprehensive review of the relevant literature is yet available, the emerging picture is that male and female ranging patterns typically vary with the type of mating system (Wittenberger 1981). Under promiscuity and some types of polygyny, males are more mobile than females; that is, they have larger home ranges or territories (Brown 1966; Lockie 1966; Ewer 1968; Trivers 1972). This seems to be an expression of male-male competi-

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tion for access to mates, since in such cases the sex difference in ranging patterns often disappears outside the breeding season (Grubb and Jewell 1966; Trivers 1972; Ferner 1974; Stamps 1977; Farentinos 1979). Under monogamy, however, ranging patterns differ little between the sexes, the norm being a shared and jointly defended home range (Blair 1951; Chivers 1974; Tenaza 1975; Kleiman 1977).

This association between mating system and the relative range sizes of males and females is not without exception. For example, in some effectively polygynous lek-breeding species, males are strongly territorial and persistently localized but females forage widely. Moreover, among highly migratory species the breeding ranges of both males and females are minute compared with the size of their annual range. This annual range, which is often very similar for the two sexes, is probably more important in selecting for navigation skills, and its effects could overshadow any consequences arising out of sex differences in breeding range. Nevertheless, these exceptions do not invalidate the model developed here, but merely define its sphere of relevance and suggest criteria for the choice of species used to evaluate it.

We take as a working assumption the idea that natural selection favors spatial ability in proportion to the amount of spatial data an animal must process in its normal movements. In nonmigratory animals we can use home-range size as a proxy measure for this amount of spatial data. These assumptions may be unrealistic if the spatial complexity of the environment differs widely for the groups being compared or if the spatial information is processed and stored in ways that are adjusted for scale, such that an equivalent amount of information is present regardless of the actual size of the area being conceptualized. The first objection is tempered by the fact that, in examining sex differences in spatial ability, we inevitably compare males and females of the same species. Since habitat use, and thus environmental complexity, overlaps considerably between males and females, the two sexes tend to serve as their own controls in this regard. Addressing the second objection requires a level of measurement of cognitive processing that is unattainable at present. In the final analysis, the value of any model and its associated assumptions depends on its predictive power, and in this respect, the model developed here performs quite well (see "Results," below).

We have suggested that ranging patterns vary with both sex and mating system, and that selection should favor spatial ability in proportion to the demands of these ranging patterns. These assumptions yield the following testable prediction: sex differences in spatial ability evolve under mating systems that select for sexually dimorphic ranging patterns (e.g., some forms of promiscuity, polygyny, and polyandry), but such cognitive sex differences will be reduced under mating systems in which the disruptive selection for range size is minimal (e.g., monogamy).

CHOICE OF SUBJECT

Many lines of evidence suggest that *Homo sapiens* is mildly polygynous (Alexander et al. 1979; Gaulin 1980; Harcourt et al. 1981; Martin and May 1981; Short

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1981; Daly and Wilson 1983; Gaulin and Boster 1985). Cross-cultural surveys indicate that polygyny is practiced in the vast majority of societies (e.g., 708 out of 849 societies catalogued in Murdock 1967). Moreover, humans exhibit a variety of sex differences that are characteristic of polygynous and promiscuous mammals, but typically absent in monogamous species. Relative to females, human males are larger (Harrison et al. 1977; Alexander et al. 1979), are more active in courtship (Daly and Wilson 1983), have higher mortality rates (Shapiro et al. 1968; Kitagawa and Hauser 1973), and exhibit a delayed onset of sexual maturity (Tanner 1962). Given this polygynous pattern, the present hypothesis leads us to expect that humans would exhibit sex differences in spatial ability. Nevertheless, the observation that they do so is hardly a convincing test of the hypothesis. Such a test critically depends on whether or not *monogamous* species exhibit sex differences in spatial ability. Unfortunately, none of the species for which spatial ability has been previously investigated is unambiguously monogamous.

The hypothesized link between mating system and relative spatial ability of the sexes can best be examined by way of a controlled comparison of male and female ranging patterns and spatial ability in species whose reproductive tactics differ but whose phylogenetic history is similar. Among the rodents, the genus *Microtus* includes several species whose mating systems differ dramatically, thus allowing such a comparison.

Meadow voles (*Microtus pennsylvanicus*) exhibit a classically promiscuous mating pattern. Both sexes are solitary, each individual exploiting its own unique home range. Males typically exploit larger ranges and compete with one another to include a maximum number of female ranges within their own range (Getz 1961; Madison 1980*a*,*b*). Pine voles (*M. pinetorum*) are social and live in monogamous groups, as well as in what may be polyandrous groups. A single breeding female shares a home range with juveniles, subadults, and adults of both sexes, including one or more fertile males (FitzGerald and Madison 1983).

In addition to these field data, laboratory findings also suggest major differences in the mating systems of these two rodents. Dewsbury (1981) has proposed a set of 11 correlates of monogamy and examined more than 40 species of rodents, ranking them according to the extent to which they exhibit these monogamous traits. Among the proposed correlates of monogamy are such diverse characters as the absence of a Coolidge effect, long latencies to copulation after the introduction of opposite-sexed adults, small litter sizes, and paternal care. For our present purposes, the important result is that, of the eight *Microtus* species studied, *M. pinetorum* compiled the highest overall monogamy score and *M. pennsylvanicus* the lowest (Dewsbury 1981, p. 154). These laboratory data accord well with the marked behavioral differences between pine and meadow voles in the wild.

MATERIALS AND METHODS

Field Study

Meadow voles and pine voles were studied in established orchards in Pennsylvania and New York, respectively, during July and August 1982. Study popula-

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tions were judged to be in active breeding condition because many females were perforate, pregnant, or lactating and most apparently adult males were scrotal. Because reproductive tactics within a single species may vary as a function of habitat (Pitelka et al. 1974; Carey and Nolan 1975; Tattersall 1976; Fricke 1980), we began by assessing the ranging patterns of our two study populations. Telemetric techniques were used because they allow the rapid accumulation of many precise positional records for each subject, and because they are free from the possible biasing effect of trap location and baiting. An attempt was made to capture all adults within the designated study areas. Into each subject was surgically implanted (interperitoneally, through an incision in the ventrolateral abdominal wall) a small transmitter package consisting of an SM-1 transmitter (AVM Instrument Co., Dublin, Calif.) and an E13E hearing-aid battery (Eveready®), encapsulated in Elvax® paraffin (Mini-Mitter Co., Indianapolis, Ind.). Each transmitter was pretuned to a different radio frequency, and, as implanted, the transmitter packages weighed between 2.4 and 2.6 g. This represented less than 5% of body weight for the largest meadow voles and approximately 10% for the smallest pine voles. Although we have no evidence on the matter, transmitter weight may have modified movement patterns. It should not have affected males and females differently, however, since in both species, sex differences in size are slight.

After recovery from anesthesia, subjects were returned to the site of capture and released. They were allowed 3 days in the field for further recovery and adjustment to the weight of the transmitter. During recovery and radio-tracking phases, traps were left in place but were unbaited and unarmed. On the fourth day following surgery, round-the-clock data collection was begun. The position of each animal was measured and recorded on scale maps of the study area once each hour for a total of 48 h. Telemetric data were digitized, and home-range areas were computed for each subject using the minimum-convex-polygon algorithm.

After the radio-tracking phase was completed, trapping was reinitiated, and all captured adults, whether carrying transmitters or not, were transported to the laboratory for further study. Thus, all animals in the spatial-ability experiments were derived from the same local populations for which ranging data were collected, but for some individuals there were no telemetric field observations.

Laboratory Study

After the transmitters were surgically removed, all animals were caged singly and maintained on a cycle of 15 h light and 9 h dark to simulate breeding-season conditions. Following a brief period of adjustment to laboratory conditions, subjects were placed on a food-restricted regimen, and spatial-ability experiments were begun. These experiments involved an initial training phase and a subsequent testing phase, following a design used by Tolman and others to measure spatial ability in rats (Tolman et al. 1946; Tolman 1948; Dawson 1972). In the training phase, subjects were given several trials per day in a maze (fig. 1*A*) where a food reward was placed at position g. To aid orientation a scent cue was located away from the reward site, at h. Subjects were trained to find the reward from progressively more-distant starting points. Thus, in the first three trials, subjects started from position f; in the next three, from e; then three trials each from d, c, and b. All subsequent trials began from position a.

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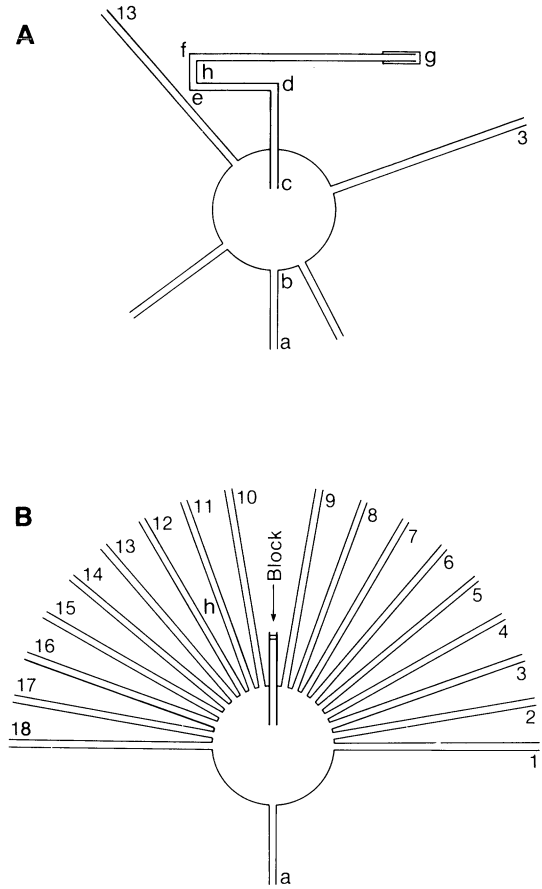


FIG. 1.—Plans of the experimental mazes. *A*, The training maze; subjects were trained to find the food reward at g from progressively more-distant points (f, e, d, etc.). *B*, The testing maze, in which the previously rewarded route is blocked and no food is present; path 6 leads directly to the place where food had been present in the training maze. In both mazes a scent cue is present at h.

The two species were trained and tested as separate experimental groups. Each group was considered to have attained acceptable levels of performance (criterion) when 40% of the subjects ran the maze perfectly (no wrong turns) from position a. Within each species, every subject ran the same number of trials in the training maze before testing; the two species differed in pretest experience, however, because the meadow vole group reached criterion levels of performance with fewer trials. The implications of this difference are discussed below.

On the day following attainment of criterion, subjects were tested in a modified maze in which the previously rewarded (entrained) path was blocked but several new alternatives were available (fig. 1*B*). One of these, path 6, led directly from the central arena to the place where the food reward had been located in the training phase. No food was present in the testing phase to prevent the subjects

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from simply using olfactory cues to locate a reward. In addition, to avoid the possibility of animals' using scent cues to follow the path taken by previous subjects, the physical runways were rinsed and redistributed randomly to positions 1 through 18 for each subject. The positioning of the training and testing mazes in the laboratory was identical, so as not to disorient subjects using extra-maze cues (O'Keefe 1978). The testing maze is used to evaluate the extent to which subjects can properly orient their movements toward an area that has consistently held a reward, when they are prevented from using a known route.

Each subject's spatial ability was ranked relative to that of all other conspecifics on the basis of two features of its performance during a single trial in the testing maze: (1) how easily it found the previously rewarded route, and (2) how appropriate an alternative it chose after discovering that the old route was blocked. The first feature was scored as the number of paths explored before locating the entrained path. Badly disoriented subjects had considerable difficulty locating the entrained route. This first feature provided crude spatial-ability ranks for the subjects, but since in both species the majority of animals located the entrained route with no more than two incorrect choices (much better than chance performance), many ties resulted. Following Tolman (1948), the second feature was used to resolve these ties. Path 6, leading directly to the previously rewarded site, was assigned a value of zero, and every other path was assigned a value based on its proximity to path 6: one for paths 5 and 7, two for paths 4 and 8, and so on. Each subject received a score on the second feature equal to the sum of the values of the first two paths chosen after discovery and rejection of the entrained path. On both features, lower scores indicate better performance. Together, the two features yielded a unique spatial-ability rank for each subject.

RESULTS

Field Study

The telemetry data strongly support prior descriptions of mating and ranging patterns in *Microtus pennsylvanicus* and *M. pinetorum*. In meadow voles, the home ranges of females overlap very little; male ranges are consistently larger, overlap somewhat, and tend to include the ranges of several females. In pine voles, social groups include a small set of individuals that share both a home range and one or more nest sites within it.

Quantitative data on ranging patterns are presented by species and sex in figure 2 and table 1. Both transmitter failure and predation reduced the number of observations (telemetric data points) per individual in the field studies. Thus, we first tested whether the number of observations differed significantly between sexes, within species. As the table shows, the slight differences that do exist are not significant (two-tailed, separate-variance *t*-tests).

We then tested for significant differences in range size using a variety of techniques. For both species the null hypothesis is that there is no sex difference in range size, and the alternative hypothesis is that male ranges are larger. We selected directional tests because in mammals, males generally have larger ranges

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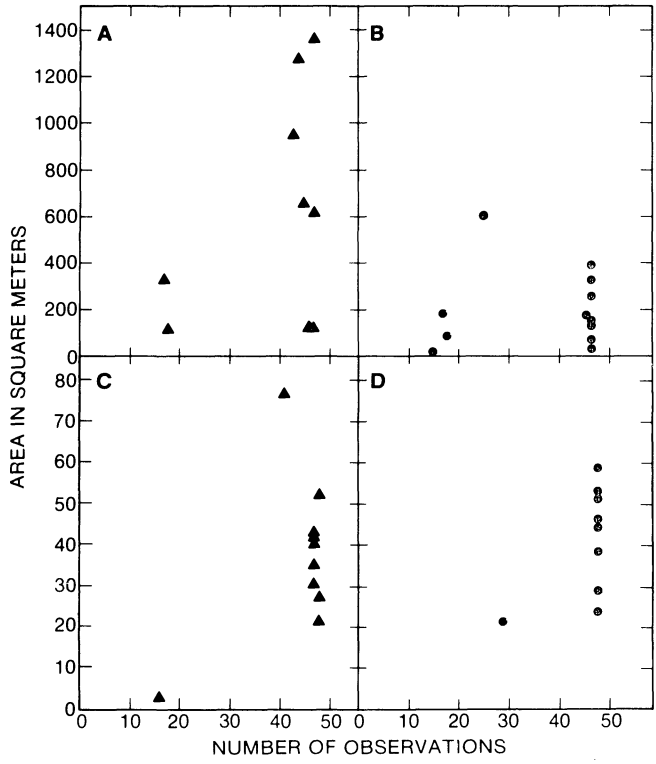


FIG. 2.—Computed home-range area by number of telemetric observations for individual voles. *A*, Male meadow voles; *B*, female meadow voles; *C*, male pine voles; *D*, female pine voles. Note that the scale on the *y*-axis is the same for both sexes within each species but different between the species.

TABLE 1

t-TESTS FOR SIGNIFICANT SEX DIFFERENCES

	OBSERVATIONS		AREA		AREA/OBSERVATION	
SPECIES	Male	Female	Male	Female	Male	Female
Meadow vole						
Mean	39.44	37.75	623.26	209.35	15.61	6.38
SD	12.53	13.62	491.07	168.16	10.27	6.44
N	9	12	9	12	9	12
P	.771		.019		.017	
Pine vole						
Mean	42.70	44.89	37.57	40.80	0.85	0.90
SD	9.60	6.33	19.56	13.42	0.46	0.24
N	10	9	10	9	10	9
P	.562		.661		.621	

NOTE.—The number of observations (telemetric data points), the home-range area, and the ratio of area to observations are tested for significant differences within species and between sexes. Tests on the number of observations are two-tailed; tests on area and area/observation are one-tailed. In no case would a change in the directionality of the test importantly alter the result.

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(Brown 1966; Lockie 1966; Ewer 1968; Trivers 1972), although none of the results would change if nondirectional tests were used. Comparisons of total home-range area indicate that, among meadow voles, males have significantly larger ranges, whereas among pine voles, females have slightly but insignificantly larger ranges (table 1; one-tailed, separate-variance t-tests).

A careful reading of the table shows that, for both species, the sex with a higher mean number of observations has the larger mean range size. Measured range size increases to some asymptote with increasing numbers of observations. Thus, one might question whether, despite the statistical insignificance of sex differences in the number of observations, these small differences could affect the results of our area tests. We evaluated this possibility in two ways, the first of which also appears in the table. For each subject, home-range area was divided by the number of observations, and this "ranging index" was examined for significant sex differences. The pattern of results is unchanged: meadow voles show a significant sex difference; pine voles do not. This technique implicitly assumes a linear (not asymptotic) relationship between range size and number of observations. The effects of violating this assumption depend on whether, for most subjects, we reached the relatively flat part of the curve of observation number versus home-range size, a matter on which we cannot give quantitative evidence. (Mares et al. [1980] suggested that 20–25 observations should be sufficient to approach the asymptote.) Thus, we performed one last, virtually assumption-free test. Only those animals for which we had between 40 and 47 observations were included (see fig. 2). We merely ranked these animals according to home-range size, using the Mann-Whitney U-test to evaluate sex differences in range size. The results hold: in meadow voles, males have significantly larger home ranges than do females ($.01 < P < .025$); for the two sexes of pine voles, we cannot reject the null hypothesis of equal range size ($P > .05$). In sum, no matter how we treat the ranging data, we cannot negate the large and significant sex difference in the effectively polygynous species, and we cannot regard the small difference in the monogamous species as anything but random.

Laboratory Study

The Mann-Whitney U-test was also used to compare rank data on the maze performance of males and females within each species (fig. 3). In meadow voles, males rank significantly higher than females on the spatial-ability task ($.01 < P < .025$). In pine voles, the maze performance of males and females does not differ significantly ($P > .05$).

Taken together, field and laboratory data from the present study support the hypothesized relationship among mating system, ranging pattern, and spatial ability. In the promiscuous meadow vole, males both range more widely than females and perform better on the maze task; in the monogamous pine vole, no sex difference is observed in either ranging pattern or spatial ability. Although our sample size is small—two species—the hypothesis has been legitimately jeopardized. The absence of a sex difference in spatial ability among pine voles is a particularly informative finding, since no other mammalian species has been shown to lack this sex difference, not even its congener, the meadow vole. This highly novel result was predicted a priori from evolutionary considerations.

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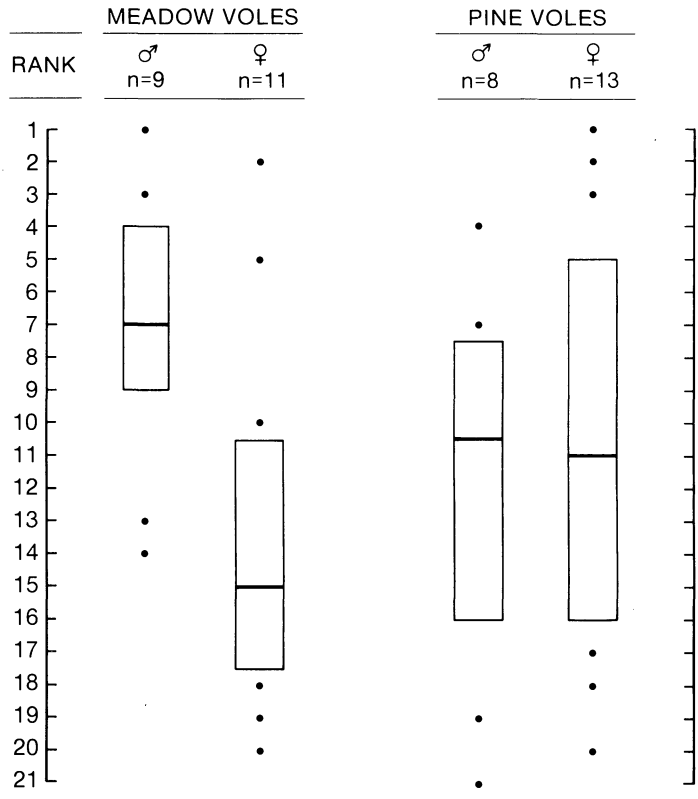


FIG. 3.—Ranked maze performance for males and females of two vole species. In each group the heavy horizontal bar represents the median rank, and the box includes the middle 50% of cases; more-deviant cases are indicated by points.

DISCUSSION

It would be valuable to know whether home-range size and spatial ability are correlated within individuals. For example, do the male meadow voles with the largest ranges also have the best maze scores? Although, for each species, all field and laboratory subjects came from a single population, a series of unfortunate circumstances (including transmitter failure, the loss of several radio-tracked animals, and relatively high mortality rates in the laboratory) combined to produce minimal overlap between our field and laboratory subject pools. Thus, the requisite correlational tests were impossible.

The present study was conceived and designed as an investigation of sex differences in spatial ability, but it nevertheless gives some insight into species differences in this trait. Independent of sex, meadow vole ranges are much larger than pine vole ranges (fig. 2). To facilitate the within-species, between-sex comparisons that were our primary objective, both species were trained to the same criterion. Thus, it is not surprising that, overall, the two species performed equally well in the testing phase. Nevertheless, a species difference in spatial

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ability is indicated by the fact that pine voles required roughly twice as many trials to reach criterion.

None of the data presented here can "resolve" the nature-versus-nurture controversy that continues to plague the spatial-ability literature. For example, if our hypothesis is accepted, it means that sexual selection shapes the ranging patterns of males and females just as it does other features of their reproductive strategies. In promiscuous and polygynous species males are expected to range more widely than females because this increased movement is an effective way of increasing their access to potential mates. Such wandering would have a much lower payoff for males in a monogamous species, in which all or most of the females encountered would already be pair-bonded. Under each type of mating system, males and females are expected to be able to solve spatial problems in proportion to the demands of their own movement patterns.

This analysis, however, fails to tell us whether males in certain promiscuous and polygynous species have higher spatial ability because they wander more and therefore get more practice manipulating spatial data, or because selection has given them higher spatial ability in order that they might wander without becoming disoriented. Were we to rear laboratory-bred "home-rangeless" voles, the highly artificial rearing environment of these animals would render their maze performance difficult to interpret. For example, even if neither species exhibited sex differences in spatial ability under such conditions, the hypothesis that selection shapes spatial ability in response to the demands of ranging patterns would not be jeopardized; selection can be expected to favor alleles that foster the development of appropriate spatial skills under normal conditions, but not necessarily under unrealistic laboratory conditions.

As a tentative conclusion we suggest that the superior spatial skills of human males are consistent with our hypothesis, and with the general pattern of polygynous traits exhibited by *Homo sapiens*. A more detailed application of this model to an analysis of human spatial ability is problematic, owing to the paucity of relevant studies. Nevertheless, at least one anthropological study can be cited in this context.

In a study of the Logoli, a polygynous, cattle-raising people of western Kenya, Munroe and Munroe (1971) measured male and female movement patterns directly. In 20 spot checks of location for age-matched pairs of boys and girls, they found that during their free time, boys were significantly farther from home than were their female age-mates. On a subsequent test of spatial ability (copying block patterns), those subjects who had been farther from home performed significantly better than their less mobile controls. These findings are consonant with the proposed link between sex, movement patterns, and spatial ability, but similar data from a variety of societies would be required for a full cross-cultural evaluation of our hypothesis.

The influence of culture aside, debate seems likely over the relative importance of evolutionary versus other more proximate factors in shaping cognitive sex differences. Such debate can be productive only with the understanding that these are complementary rather than competing levels of explanation.

The following is offered as a possible causal scenario that emphasizes integra-

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tion among various levels of explanation. A case can be made that, in humans, high spatial ability is associated with strong hemispheric lateralization (Waber 1976, 1977; McGee 1979), and that lateralization is weakest in early developers of both sexes (although there are some complex interactions with handedness and reasoning ability; Harshman and Hampson 1983). Developmental rate, however, is correlated with sex; human males typically develop later and therefore tend to have superior spatial ability (Waber 1976, 1977). From an evolutionary perspective, delayed maturation, growth to a larger adult size, and larger home ranges are all predicted for males in promiscuous and polygynous species, but not in monogamous ones (Orians 1969; Trivers 1972). If general, the ontogenetic linking of developmental rate, lateralization, and spatial ability could provide a proximate developmental mechanism underlying the emergence of sex differences in spatial ability in any nonmonogamous species. Thus, the potential exists to unite proximate and evolutionary analyses and simultaneously explain when, how, and why sex differences in spatial ability occur.

The results reported here support the emerging view that sex itself is not the key independent variable shaping "sex differences" (Williams 1966; Trivers 1972). Rather, the mating system and the associated pattern of reproductive tactics employed by males and females determine the evolutionary importance of various traits for each sex. When the optimal value of a trait does differ by sex, selection favors facultative (*sensu* Williams 1966) control mechanisms that lead to the differential expression of the trait in males and females. Thus, during ontogeny, the different internal and external environments of males and females mediate alternative expressions of relatively similar underlying genotypes. The present study demonstrates the predictive power of this evolutionary view of sex differences and extends its domain into the realm of cognitive traits.

SUMMARY

A comparison of male and female home-range size and maze performance in two congeneric species of microtine rodents suggests an evolutionary explanation for sex differences in spatial ability. Male/female differences in this cognitive trait seem to depend ultimately not on sex per se, but rather on the type of mating system and on the associated reproductive tactics employed by each sex. Through a combined field and laboratory study, the hypothesis that males evolve superior spatial ability only when they are more mobile (i.e., have larger home ranges) than females is developed and tested.

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